

ENVIRONMENTAL PRODUCT DECLARATION

IN ACCORDANCE WITH EN 15804+A2 & ISO 14025 / ISO 21930

Retrofit Thermal Absorber

Copa Solar AB



EPD HUB, HUB-1392

Published on 17.05.2024, last updated on 17.05.2024, valid until 17.05.2029

GENERAL INFORMATION

MANUFACTURER

Manufacturer	Copa Solar AB
Address	Stålgatan 15A 19572 Rosersberg
Contact details	info@copasolar.com
Website	www.copasolar.com

EPD STANDARDS, SCOPE AND VERIFICATION

Program operator	EPD Hub, hub@epdhub.com
Reference standard	EN 15804+A2:2019 and ISO 14025
PCR	EPD Hub Core PCR version 1.1, 5 Dec 2023
Sector	Construction product
Category of EPD	Third party verified EPD
Parent EPD number	
Scope of the EPD	Cradle to gate with modules C1-C4, D
EPD author	Andreas Pettersson
EPD verification	Independent verification of this EPD and data, according to ISO 14025: ☐ Internal verification ☒ External verification
EPD verifier	Imane Uald Iamkaddam, as an authorized verifier acting for EPD Hub Limited

The manufacturer has the sole ownership, liability, and responsibility for the EPD. EPDs within the same product category but from different programs may not be comparable. EPDs of construction products may not be comparable if they do not comply with EN 15804 and if they are not compared in a building context.

PRODUCT

Product name	Retrofit Thermal Absorber
Additional labels	Absorber plate Mesh 1350W
Product reference	RAM00202
Place of production	SSAB Braunschweig, Germany
Period for data	Jan. 2023 to Dec. 2023
Averaging in EPD	No averaging
Variation in GWP-fossil for A1-A3	0 %

ENVIRONMENTAL DATA SUMMARY

Declared unit	1 m2 of Retrofit Thermal
Declared unit mass	2.48 kg
GWP-fossil, A1-A3 (kgCO2e)	2,11E+01
GWP-total, A1-A3 (kgCO2e)	2,00E+01
Secondary material, inputs (%)	2.84
Secondary material, outputs (%)	122
Total energy use, A1-A3 (kWh)	107
Net fresh water use, A1-A3 (m3)	0.11

PRODUCT AND MANUFACTURER

ABOUT THE MANUFACTURER

Copa Solar AB is an absorber and PVT designer company located in Rosersberg, Sweden, in the heart of the Stockholm North Logistic Center. Copa Solar strives for excellence in environmental technologies in the solar energy area thru research and development. The contracted European manufacturing ensures the highest quality of the products towards our customers. By combining the thermal energy harvesting from construction and electrical panels we optimize the output to be the most renewable and reliable source with a long and secure lifespan.

PRODUCT DESCRIPTION

RAM00202 is a Thermal Heat Absorber plate for Retrofit. It consists of aluminium pipes on an aluminium cooling plate, prepared with adhesive stripes for mounting on the backside of PV-panels for conversion to PVT-panels. Other suitable surfaces are facadepanels, stone sheets, metal sheets and other constructions suited for thermal harvesting and cooling methodology.

The Retrofit Absorber is designed to lead away excess heat that is built up in for example PV-panels and its surroundings from the solar irradiance and makes the PV-panel work at it's optimal temperature.

The thermal energy can be used in various applications and the electrical energy production will increase due to the optimization of the panels operating temperature.

RAM00202 has a thermal power output of 1396 W at ambient temperature of 45°C and mean thermal fluid temperature of 5°C with optical efficiency of 0,53 as well as heat loss coefficients of $a_1=22,5 \text{ W/m}^2\cdot\text{K}$ and $a_2=0 \text{ W/m}^2\cdot\text{K}^2$. With a weight of 2,86 kg fully filled with thermal liquid it is compatible with the most known mounting systems. The dimensions of a RAM00202 are 1074x821x15 mm included the mounted absorber with an aperture and gross area of 0,88 m². The removal of the thermal energy is

beneficial to the electric solar panel when cooled to optimal electrical producing temperature.

Further information can be found at www.copasolar.com.

PRODUCT RAW MATERIAL MAIN COMPOSITION

Raw material category	Amount, mass- %	Material origin
Metals	100	EU, Germany
Minerals	0	N/N
Fossil materials	0	N/N
Bio-based materials	0	N/N

BIOGENIC CARBON CONTENT

Product's biogenic carbon content at the factory gate

Biogenic carbon content in product, kg C	0
Biogenic carbon content in packaging, kg C	0.396

FUNCTIONAL UNIT AND SERVICE LIFE

Declared unit	1 m2 of Retrofit Thermal
Mass per declared unit	2.48 kg
Functional unit	
Reference service life	25

SUBSTANCES, REACH - VERY HIGH CONCERN

The product does not contain any REACH SVHC substances in amounts greater than 0,1 % (1000 ppm).

PRODUCT LIFE-CYCLE

SYSTEM BOUNDARY

This EPD covers the life-cycle modules listed in the following table.

Product stage			Assembly stage		Use stage								End of life stage				Beyond the system boundaries	
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7		C1	C2	C3	C4	D	
x	x	x	MND	MND	MND	MND	MND	MND	MND	MND	MND		x	x	x	x	x	
Raw materials	Transport	Manufacturing	Transport	Assembly	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use		Deconstr./demol.	Transport	Waste processing	Disposal	Reuse	Recovery
																		Recycling

Modules not declared = MND. Modules not relevant = MNR.

MANUFACTURING AND PACKAGING (A1-A3)

The environmental impacts considered for the product stage cover the manufacturing of raw materials used in the production as well as packaging materials and other ancillary materials. Also, fuels used by machines, and handling of waste formed in the production processes at the manufacturing facilities are included in this stage. The study also considers the material losses occurring during the manufacturing processes as well as losses during electricity transmission.

A1 – Raw Material and Electricity Supply. Raw material extraction and processing are in this module taken into account when processing the material and energy during the manufacturing process. Aluminium is the main raw material for manufacturing. No specific data of recycled content is declared.

A2 – Transport to Manufacturer Transport to the manufacturing site. The

modelling includes onsite electric forklift running on no carbon electricity and a local conveyor belt and so forth the impact is also included in A3.

A3 – Manufacturing. Manufacturing, including the provision of total materials for production and the assembly of RAM00202 with an energy use of 8,064MJ per RAM00202. This module is considering all aspects of the material and has no waste to include as the absorber panel are 100% recyclable.

TRANSPORT AND INSTALLATION (A4-A5)

Transportation impacts occurred from final products delivery to construction site (A4) cover fuel direct exhaust emissions, environmental impacts of fuel production, as well as related infrastructure emissions.

All transport impacts are related to the customers

PRODUCT USE AND MAINTENANCE (B1-B7)

This EPD does not cover the use phase.

Air, soil, and water impacts during the use phase have not been studied.

PRODUCT END OF LIFE (C1-C4, D)

C1, By the end-of-life deconstruction is assumed to be similar to installation so, the hybrid solar plant deconstruction is expected to consume mainly electricity that is assumed to be equal to installation electricity (2.2 kWh/module) with zero carbon impact.

C2, A 50 km distance from hybrid solar plant to waste recycling plant was considered during downstream process. Transport was assumed to be by road (Transport, freight, lorry 16-32 metric ton, euro4 {RER}| market for transport, freight, lorry 16-32 metric ton, EURO4 | Cut-off, U).

C3, The same electricity is expected to be consumed during the dismantling stage of the module that the used for manufacturing as mainly assembly operations were done.

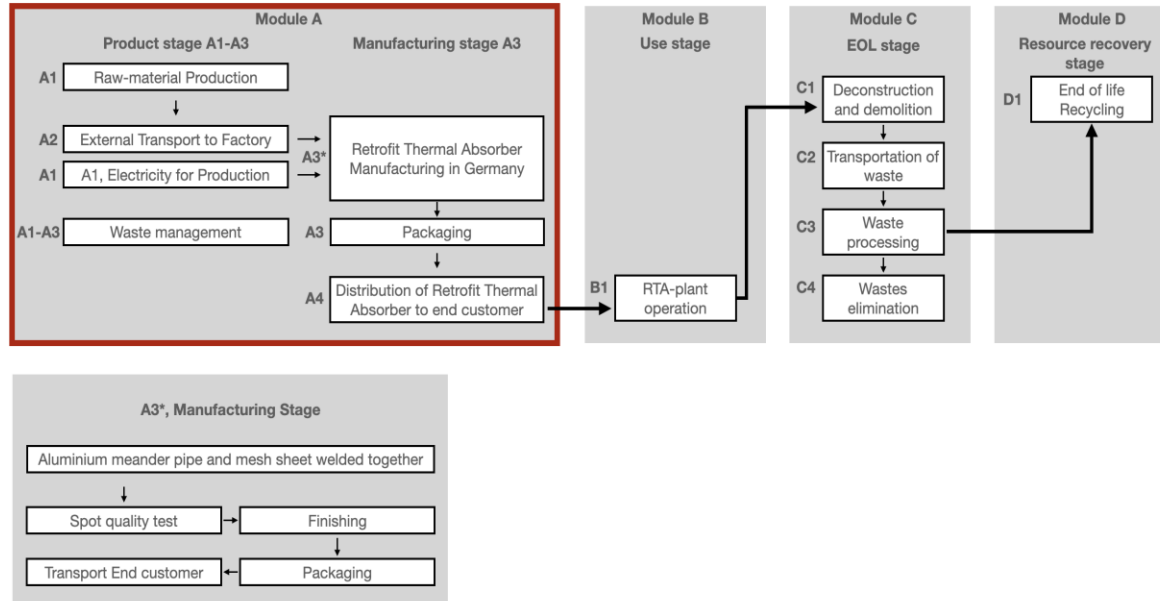
C4, In this module, the landfill percentages stated in the scenario declared in C3 are considered, based on the following scenario: The following end of life scenario was considered. Aluminium end-of-life scenario is 96%

recycled and the remaining 4 % is landfilled. Adhesive materials are shredded and a 95% is considered municipal solid waste for energy recovery and 5% considered at loss in the recycling process.

D, Potential Reuse, Recover, Recycle, module enables full recovery of material from applications that have reached End-Of-Life into the production for full recovery. The 100% Aluminium absorber may be reapplied to another panel for Reuse or sent for 100% recycling.

SYSTEM BOUNDARIES

System boundaries Retrofit Thermal Absorber (RTA)



LIFE-CYCLE ASSESSMENT

CUT-OFF CRITERIA

The study does not exclude any modules or processes which are stated mandatory in the reference standard and the applied PCR. The study does not exclude any hazardous materials or substances. The study includes all major raw material and energy consumption. All inputs and outputs of the unit processes, for which data is available for, are included in the calculation. There is no neglected unit process more than 1% of total mass or energy flows. The module specific total neglected input and output flows also do not exceed 5% of energy usage or mass.

ALLOCATION, ESTIMATES AND ASSUMPTIONS

Allocation is required if some material, energy, and waste data cannot be measured separately for the product under investigation. All allocations are done as per the reference standards and the applied PCR. In this study, allocation has been done in the following ways:

Data type	Allocation
Raw materials	No allocation
Packaging materials	Allocated by mass or volume
Ancillary materials	Allocated by mass or volume
Manufacturing energy and waste	Allocated by mass or volume

AVERAGES AND VARIABILITY

Type of average	No averaging
Averaging method	Not applicable
Variation in GWP-fossil for A1-A3	0 %

This EPD is product and factory specific and does not contain average calculations.

LCA SOFTWARE AND BIBLIOGRAPHY

This EPD has been created using One Click LCA EPD Generator. The LCA and EPD have been prepared according to the reference standards and ISO 14040/14044. The EPD Generator uses Ecoinvent v3.8, Plastics Europe, Federal LCA Commons and One Click LCA databases as sources of environmental data.

ENVIRONMENTAL IMPACT DATA

CORE ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2, PEF

Impact category	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
GWP – total ¹⁾	kg CO ₂ e	2,12E+01	2,54E-02	-1,20E+00	2,00E+01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,22E-02	1,43E+00	1,57E-01	-1,04E+01
GWP – fossil	kg CO ₂ e	2,08E+01	2,54E-02	3,29E-01	2,11E+01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,22E-02	5,54E-02	4,21E-03	-1,01E+01
GWP – biogenic	kg CO ₂ e	0,00E+00	0,00E+00	-1,53E+00	-1,53E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	1,38E+00	1,53E-01	0,00E+00
GWP – LULUC	kg CO ₂ e	3,81E-01	9,47E-06	2,16E-03	3,83E-01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	4,50E-06	7,89E-05	2,04E-06	-2,60E-01
Ozone depletion pot.	kg CFC ₁₁ e	1,69E-06	5,85E-09	4,07E-08	1,73E-06	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,80E-09	6,38E-09	1,21E-09	-1,12E-06
Acidification potential	mol H ⁺ e	1,61E-01	1,19E-04	2,39E-03	1,64E-01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	5,16E-05	6,56E-04	3,44E-05	-6,86E-02
EP-freshwater ²⁾	kg Pe	0,00E+00	2,03E-07	2,88E-05	2,90E-05	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	9,98E-08	3,29E-06	5,60E-08	-5,54E-04
EP-marine	kg Ne	1,90E-02	3,81E-05	5,89E-04	1,97E-02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,53E-05	1,36E-04	1,51E-05	-7,63E-03
EP-terrestrial	mol Ne	2,14E-01	4,20E-04	6,36E-03	2,21E-01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,69E-04	1,57E-03	1,40E-04	-8,43E-02
POCP (“smog”) ³⁾	kg NMVOCe	7,10E-02	1,29E-04	2,23E-03	7,34E-02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	5,41E-05	4,33E-04	4,20E-05	-3,21E-02
ADP-minerals & metals ⁴⁾	kg Sbe	1,22E-04	6,48E-08	1,26E-05	1,34E-04	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,86E-08	6,61E-06	2,20E-08	-2,57E-05
ADP-fossil resources	MJ	1,63E+02	3,81E-01	4,45E+00	1,68E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,83E-01	7,98E-01	9,70E-02	-1,59E+02
Water use ⁵⁾	m ³ e depr.	8,20E+00	1,71E-03	3,44E-01	8,54E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	8,19E-04	1,65E-02	2,24E-03	-2,12E+01

1) GWP = Global Warming Potential; 2) EP = Eutrophication potential. Required characterisation method and data are in kg P-eq. Multiply by 3,07 to get PO₄e; 3) POCP = Photochemical ozone formation; 4) ADP = Abiotic depletion potential; 5) EN 15804+A2 disclaimer for Abiotic depletion and Water use and optional indicators except Particulate matter and Ionizing radiation, human health. The results of these environmental impact indicators shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.

ADDITIONAL (OPTIONAL) ENVIRONMENTAL IMPACT INDICATORS – EN 15804+A2, PEF

Impact category	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Particulate matter	Incidence	1,65E-06	2,94E-09	7,36E-08	1,72E-06	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,40E-09	8,25E-09	7,02E-10	-8,01E-07
Ionizing radiation ⁶⁾	kBq U235e	1,23E+00	1,85E-03	4,52E-02	1,27E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	8,72E-04	1,10E-02	1,02E-03	-2,91E+00
Ecotoxicity (freshwater)	CTUe	5,06E+02	3,38E-01	1,28E+01	5,19E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,65E-01	3,08E+00	1,54E+02	-1,90E+02
Human toxicity, cancer	CTUh	5,14E-08	9,72E-12	5,20E-10	5,19E-08	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	4,05E-12	9,60E-11	7,87E-12	6,99E-09
Human tox. non-cancer	CTUh	7,89E-07	3,48E-10	1,36E-08	8,03E-07	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,63E-10	4,19E-09	1,55E-10	-4,65E-07
SQP ⁷⁾	-	2,36E+01	4,09E-01	1,94E+02	2,18E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,11E-01	1,35E+00	2,12E-01	-1,58E+02

6) EN 15804+A2 disclaimer for ionizing radiation, human health. This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator; 7) SQP = Land use related impacts/soil quality.

USE OF NATURAL RESOURCES

Impact category	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Renew. PER as energy ⁸⁾	MJ	1,09E+02	4,49E-03	2,65E+01	1,35E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,06E-03	1,41E-01	1,46E-02	-8,18E+01
Renew. PER as material	MJ	0,00E+00	0,00E+00	1,24E+01	1,24E+01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	-1,12E+01	-1,24E+00	0,00E+00
Total use of renew. PER	MJ	1,09E+02	4,49E-03	3,90E+01	1,48E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,06E-03	-1,11E+01	-1,23E+00	-8,18E+01
Non-re. PER as energy	MJ	2,44E+02	3,81E-01	4,45E+00	2,49E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,83E-01	7,98E-01	9,71E-02	-1,59E+02
Non-re. PER as material	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Total use of non-re. PER	MJ	2,44E+02	3,81E-01	4,45E+00	2,49E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,83E-01	7,98E-01	9,71E-02	-1,59E+02
Secondary materials	kg	7,05E-02	1,09E-04	7,87E-03	7,85E-02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	5,08E-05	7,88E-04	1,40E-04	1,74E+00
Renew. secondary fuels	MJ	0,00E+00	1,12E-06	1,53E-04	1,54E-04	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	5,13E-07	3,85E-05	1,38E-06	-3,04E-04
Non-ren. secondary fuels	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Use of net fresh water	m ³	1,04E-01	4,90E-05	7,80E-03	1,12E-01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,37E-05	5,02E-04	1,34E-04	-4,61E-01

8) PER = Primary energy resources.

END OF LIFE – WASTE

Impact category	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Hazardous waste	kg	5,87E+00	4,91E-04	3,63E-02	5,91E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,43E-04	5,06E-03	0,00E+00	-2,23E+00
Non-hazardous waste	kg	3,69E+01	8,17E-03	7,29E-01	3,76E+01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	3,99E-03	1,74E-01	3,37E-01	-2,47E+01
Radioactive waste	kg	9,81E-04	2,56E-06	1,91E-05	1,00E-03	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,22E-06	4,85E-06	0,00E+00	-1,01E-03

END OF LIFE – OUTPUT FLOWS

Impact category	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Components for re-use	kg	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Materials for recycling	kg	0,00E+00	0,00E+00	7,30E-02	7,30E-02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	3,02E+00	0,00E+00	0,00E+00
Materials for energy rec	kg	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Exported energy	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	0,00E+00	0,00E+00	0,00E+00	0,00E+00

ENVIRONMENTAL IMPACTS – EN 15804+A1, CML / ISO 21930

Impact category	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Global Warming Pot.	kg CO ₂ e	2,07E+01	2,51E-02	3,22E-01	2,10E+01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,21E-02	5,46E-02	9,82E-03	-1,01E+01
Ozone depletion Pot.	kg CFC ₁₁ e	1,66E-06	4,64E-09	3,59E-08	1,70E-06	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,22E-09	5,17E-09	9,61E-10	-9,54E-07
Acidification	kg SO ₂ e	1,35E-01	9,11E-05	1,91E-03	1,37E-01	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	4,01E-05	5,31E-04	2,55E-05	-5,92E-02
Eutrophication	kg PO ₄ ³ e	3,66E-02	2,10E-05	8,35E-04	3,74E-02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	9,13E-06	1,89E-04	2,58E-04	-2,21E-02
POCP ("smog")	kg C ₂ H ₄ e	1,06E-02	3,36E-06	1,73E-04	1,08E-02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,57E-06	2,02E-05	2,30E-06	-6,27E-03
ADP-elements	kg Sbe	1,58E-04	6,29E-08	1,26E-05	1,71E-04	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	2,77E-08	6,60E-06	2,05E-08	-2,30E-05
ADP-fossil	MJ	2,44E+02	3,81E-01	4,45E+00	2,49E+02	MND	MND	MND	MND	MND	MND	MND	MND	MND	MNR	1,83E-01	7,97E-01	9,70E-02	-1,59E+02

VERIFICATION STATEMENT

VERIFICATION PROCESS FOR THIS EPD

This EPD has been verified in accordance with ISO 14025 by an independent, third-party verifier by reviewing results, documents and compliance with reference standard, ISO 14025 and ISO 14040/14044, following the process and checklists of the program operator for:

- This Environmental Product Declaration
- The Life-Cycle Assessment used in this EPD
- The digital background data for this EPD

Why does verification transparency matter? [Read more online](#)

This EPD has been generated by One Click LCA EPD generator, which has been verified and approved by the EPD Hub.

THIRD-PARTY VERIFICATION STATEMENT

I hereby confirm that, following detailed examination, I have not established any relevant deviations by the studied Environmental Product Declaration (EPD), its LCA and project report, in terms of the data collected and used in the LCA calculations, the way the LCA-based calculations have been carried out, the presentation of environmental data in the EPD, and other additional environmental information, as present with respect to the procedural and methodological requirements in ISO 14025:2010 and reference standard.

I confirm that the company-specific data has been examined as regards plausibility and consistency; the declaration owner is responsible for its factual integrity and legal compliance.

I confirm that I have sufficient knowledge and experience of construction products, this specific product category, the construction industry, relevant standards, and the geographical area of the EPD to carry out this verification.

I confirm my independence in my role as verifier; I have not been involved in the execution of the LCA or in the development of the declaration and have no conflicts of interest regarding this verification.

Imane Uald lamkaddam, as an authorized verifier acting for EPD Hub Limited
17.05.2024

